An Introduction to Chaotic Dynamical Systems

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Robert L. Devaney
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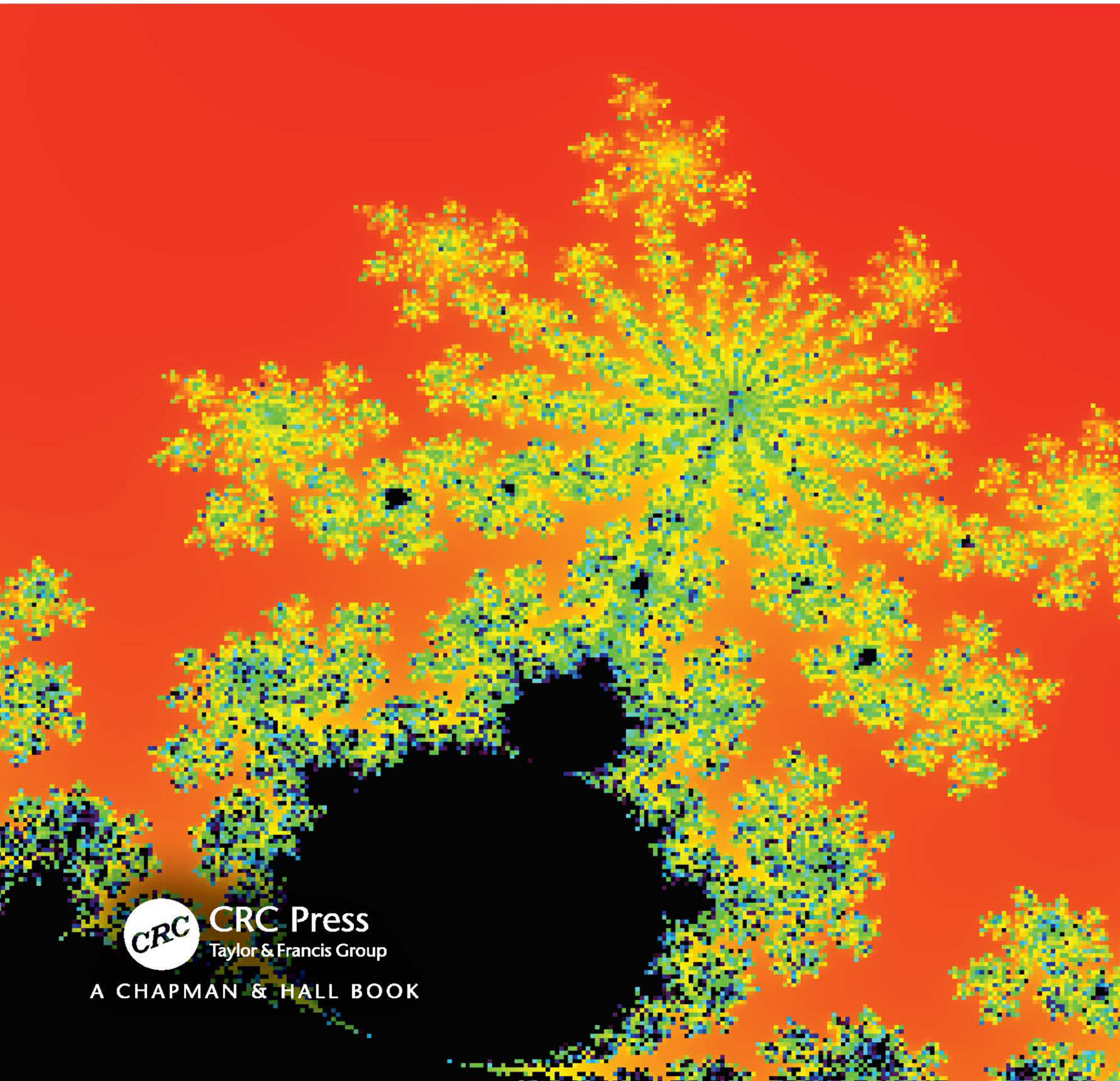
AN INTRODUCTION TO CHAOTIC DYNAMICAL SYSTEMS

Third Edition

Robert L. Devaney



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Preface to the Third Edition

In the thirty years since the previous edition of this text appeared, many amazing things have occurred in the field of mathematics known as dynamical systems. Indeed, there has been an explosion of interest in this field in the mathematical community as well as in many areas of science. Scientists and engineers have come to realize the power and the beauty of the geometric and qualitative techniques developed during this period. More importantly, they have been able to apply these techniques to a number of important nonlinear problems ranging from physics and chemistry to ecology and economics. The results have been truly exciting: systems that once seemed completely intractable from an analytic point of view can now be understood in a geometric or qualitative sense rather easily. Chaotic and random behavior of solutions of deterministic systems is now understood to be an inherent feature of many nonlinear systems, and the geometric theory developed over the past few decades handles this situation quite nicely.

Perhaps the most important development in the past thirty years has been the widespread availability of computers for use in the study of dynamical systems. Computer graphics have played an extremely important role here. Now, rather than finding specific solutions to dynamical systems (which is almost always impossible), computer graphics have allowed us to view the dynamical behavior geometrically. This has led to a major new approach to the study of dynamical systems. In addition, the appearance of incredibly beautiful and intricate objects such as the Mandelbrot set, the Julia set, and other fractals have really piqued interest in the field.

There are many different types of mathematical dynamical systems, including ordinary differential equations, partial differential equations, ergodic systems, and discrete dynamical systems. In this book, we will concentrate only on discrete dynamical systems (basically, iteration of mathematical functions). This approach has the advantage of making many of the ideas from other types of dynamical systems much more accessible. Indeed, the aim of this text is to make the techniques in dynamical systems available to advanced mathematics majors as well as to graduate students and scientists in other disciplines.

The field of dynamical systems and especially the study of chaotic systems has been hailed as one of the important breakthroughs in science in this century. While the field is still relatively young, there is no question that the field is becoming more and more important in a variety of scientific disciplines.

We hope that this text serves to excite and to lure many others into this "dynamic" field.

A NOTE TO THE READER:

This is, first of all, a Mathematics text that is aimed primarily at advanced undergraduate and beginning graduate students in mathematics. Throughout, we emphasize the mathematical aspects of the theory of discrete dynamical systems, not the many and diverse applications of this theory. The text begins at a relatively unsophisticated level in Chapter 1, where we deal with one-dimensional dynamics (primarily iteration of quadratic functions on the real line). This part of the text is accessible to students with only a solid background in freshman calculus. As we proceed, more advanced topics such as topology, complex analysis, and linear algebra become important. Many of these prerequisites are presented in the appendix.

The first chapter, one-dimensional dynamics, is by far the longest. It is the author's belief that virtually all of the important ideas and techniques of nonlinear dynamics can be introduced in the setting of the real line or the circle. This has the obvious advantage of minimizing the topological complications of the system and the algebraic machinery necessary to handle them. In particular, the only real prerequisite for this chapter is a good calculus course. (Oh well, we do multiply a 2 x 2 matrix once or twice in Section 13, and we use the Implicit Function Theorem in two variables in Section 12, but these are exceptions.) With only these tools, we manage to introduce such important topics as chaos, structural stability, topological conjugacy, the shift map, homoclinic points, and bifurcation theory. To emphasize the point that chaotic dynamics occur in the simplest of systems, we carry out most of our analysis in this section on two basic models, the quadratic mappings given by $F\mu(x) = \mu x(l - x)$ or $Q_c(x) = x^2 + c$. These maps have the advantage of being perhaps the simplest nonlinear maps, yet they illustrate virtually every concept we wish to introduce. A few topological ideas, such as the notion of a dense set or a Cantor set, are introduced in detail when needed.

Just to emphasize the contemporary nature of the field of discrete dynamical systems, because of the chaotic behavior of these quadratic maps, we finally understood the total behavior of these maps in the 1990s. Think about this: mathematicians finally understood $x^2 + c$ around thirty years ago! And, to this day, we still don't completely understand complex $z^2 + c$!

In previous editions of this book, the topic of complex dynamics was covered in the third and final chapters. Given the amazing resurgence of interest in this field due to the spectacular computer graphics images, we have greatly expanded coverage of this topic and moved it to Chapter 2. We again concentrate primarily on quadratic functions in the complex plane, specifically $Q_c(z) = z^2 + c$. This is the function whose overall behavior is summarized in the beautiful and intricate image known as the Mandelbrot set. Moving to the complex plane allows us to view the real quadratic functions in a very different and much more helpful way. In this chapter, we also introduce some very different dynamical behavior that arises when complex functions such as rational maps and the exponential are itereated. This chapter assumes that students are familiar with the complex plane as well as complex arithmetic and geometry. Some topics from the elementary complex analysis are used, and these are presented in the appendix.

The third chapter is devoted to higher dimensional dynamical systems. With many of the prerequisites already introduced in the first chapter, the discussion of such higher dimensional maps as Smale's horseshoe, hyperbolic toral automorphisms, and the solenoid become especially simple. This chapter assumes that the reader is familiar with some multi-dimensional calculus as well as linear algebra, including the notion of eigenvalues and eigenvectors for 3 x 3 matrices and some additional concepts from topology. All of these prerequisites are described in the appendix. One of the major differences between one dimensional and higher dimensional dynamics, the possibility of both contraction and expansion at the same time, is treated at length in a section devoted to the proof of the Stable and Unstable Manifold Theorem. We end the chapter with a lengthy set of exercises all centered on the important Henon map of the plane. This section serves as a summary of many of the previous topics in the book as well as a good "final" project for the reader. Chapters 2 and 3 are independent of one another; they both only assume familiarity with the basic concepts of dynamics as presented in the first chapter. We also have provided some background mathematical material from calculus, linear algebra, geometry and topology, and complex analysis for use in these chapters in the appendix.

There are many themes developed in this book. We have tried to present several different dynamical concepts in their most elementary formulation in Chapter 1 and to return to these subjects for further refinement at later stages in the book. One such topic is bifurcation theory. We introduce the most elementary bifurcations, the saddle-node, and the period-doubling bifurcations, early in Chapter l. Later in the same chapter, we treat the accumulation points of such bifurcations which occur when a homoclinic point develops. In Chapter 2, we revisit these topics, but from a very different perspective of the complex plane. We explore the very different bifurcations that occur in complex dynamics and include a discussion of the global aspects of the saddle-node bifurcation, and the exploding Julia sets that occur for the exponential map. In Chapter 3, we return to bifurcation theory to discuss the Hopf bifurcation. Another recurrent theme throughout the book is symbolic dynamics. We think of symbolic dynamics as a tool whereby complicated dynamical systems are equated with seemingly quite different systems, but these systems have the advantage that they can be analyzed much more easily. Symbolic dynamics appears quite early in Chapter 1 when we first describe the chaotic behavior of the quadratic map. It is clear that the most elementary setting for the phenomena associated with the Smale horseshoe mapping occurs in one dimension, and we fully exploit this idea. Later, symbolic dynamics is extended to the case of subshifts of finite type via another quadratic example. And finally, the related concepts of Markov partitions and inverse limits are introduced in the third chapter.

Examples abound in this text. We often motivate new concepts by working through them in the setting of a specific dynamical system. In fact, we have often sacrificed generality in order to concentrate on a specific system or class of systems. Many of the results throughout the text are stated in a form that is nowhere near full generality. We feel that the general theory is best left to more advanced texts, which presuppose much more advanced mathematics.

Much of what many researchers consider dynamical systems has been deliberately left out of this text. For example, we do not treat continuous systems such as differential equations at all. There are several reasons for this. First, as is well known, computations with specific nonlinear ordinary differential equations are next to impossible. Secondly, the study of differential equations necessitates a much higher level of sophistication on the part of the student, certainly more than that necessary for Chapter 1 of this text. We adopt instead the attitude that any dynamical phenomena that occur in a continuous system also occurs in a discrete system, and so we might as well make life easy and study iterated maps first. There are many texts currently available that treat continuous systems almost exclusively. We hope that this book presents a solid introduction to the topics treated in these more advanced texts.

Another topic that has been excluded is ergodic theory. It is our feeling that measure theory would take us too far afield in this book. Of course, it can be argued that measure theory is no more advanced than the complex analysis necessary for Chapter 2. However, we feel that the topological approach adopted throughout this text is inherently easier to understand, at least for undergraduates in Mathematics. There is no question, however, that ergodic theory would provide an ideal sequel to the material presented here, as would a course in nonlinear differential equations.

Finally, I am indebted to Bob Ross at CRC Press for arranging the publication of the long-overdue third edition of this text.

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April, 2021

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Robert L. Devaney is currently a Professor of Mathematics at Boston University. He received his PhD from the University of California at Berkeley under the direction of Stephen Smale. He taught at Northwestern University and Tufts University before coming to Boston University in 1980. His main area of research is dynamical systems, primarily complex analytic dynamics, but also including more general ideas about chaotic dynamical systems. Lately, he has become intrigued with the incredibly rich topological aspects of dynamics, including such things as indecomposable continua, Sierpinski curves, and Cantor bouquets. He is also the author of *A First Course in Chaotic Dynamical Systems*, Second Edition, published by CRC Press.

Part I

One Dimensional Dynamics

1

A Visual and Historical Tour

Rather than jump immediately into the mathematics of dynamical systems, we will begin with a brief tour of some of the amazing computer graphics images that arise in this field. One of our goals in this book is to explain what these images mean, how they are generated on the computer, and why they are important in mathematics. We will do none of the mathematics in this chapter. For now, you should simply enjoy the images. We hope to convince you, in the succeeding chapters, that the mathematics behind these images is even prettier than the pictures. In the second part of this chapter, we will present a brief history of some of the developments in dynamical systems over the past century. You will see that many of the ideas in dynamics arose fairly recently. Indeed, none of the computer graphics images from the tour had been seen before 1980!

1.1 Images from Dynamical Systems

This book deals with some very interesting, exciting, and beautiful topics in mathematics—topics which, in many cases, have been discovered only in the last fifty years. The main subject of the book is *dynamical systems,* the branch of mathematics that attempts to understand processes in motion. Such processes occur in all branches of science. For example, the motion of the planets and the galaxies in the heavens is a dynamical system, one that has been studied for centuries by thousands of mathematicians and scientists. The stock market is another system that changes in time, as does the earth's weather. The changes chemicals undergo, the rise and fall of populations, and the motion of a simple pendulum are classical examples of dynamical systems in chemistry, biology, and physics. Clearly, dynamical systems abound.

What does a scientist wish to do with a dynamical system? Well, since the system is moving or changing in time, the scientist would like to predict where the system is heading, where it will ultimately go. Will the stock market go up or down? Will it be rainy or sunny tomorrow? Will these two chemicals explode if they are mixed in a test tube?

Clearly, some dynamical systems are predictable, whereas others are not. You know that the sun will rise tomorrow and that when you add cream to

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a cup of coffee, the resulting "chemical" reaction will not be an explosion. On the other hand, predicting the weather a month from now or the Dow Jones average a week from now seems impossible. You might argue that the reason for this unpredictability is that there are simply too many variables present in meteorological or economic systems. That is indeed true in these cases, but this is by no means the complete answer. One of the remarkable discoveries of twentieth-century mathematics is that very simple systems, even systems depending on only one variable, may behave just as unpredictably as the stock market, just as wildly as a turbulent waterfall, and just as violently as a hurricane. The culprit, the reason for this unpredictable behavior, has been called "chaos" by mathematicians.

Because chaos has been found to occur in the simplest of systems, scientists may now begin to study unpredictability in its most basic form. It is to be hoped that the study of these simpler systems will eventually allow scientists to find the key to understanding the turbulent behavior of systems involving many variables such as weather or economic systems.

In this book we discuss the chaos in these simple settings. We will see that chaos occurs in elementary mathematical objects—objects as familiar as quadratic functions—when they are regarded as dynamical systems. You may feel at this point that you know all there is to know about quadratic functions—after all, they are easy to evaluate and to graph. You can differentiate and integrate them. But the keywords here are "dynamical systems." We will treat simple mathematical operations like taking the square root, squaring, or cubing as dynamical systems by repeating the procedure over and over, using the output of the previous operation as the input for the next. This process is called *iteration*. This procedure generates a list of real or complex numbers or points in a higher dimensional space that are changing as we proceed—this is our dynamical system. Sometimes we will find that, when we input certain numbers into the process, the resulting behavior is completely predictable, while other inputs yield results that are often bizarre and totally unpredictable.

For the types of complex functions that we will consider later, the set of numbers that yield chaotic or unpredictable behavior in the plane is called the *Julia set* after the French mathematician Gaston Julia, who first formulated many of the properties of these sets in the 1920s. These Julia sets are spectacularly complicated, even for quadratic functions. They are examples of *fractals*. These are sets which, when magnified over and over again, always resemble the original image. The closer you look at a fractal, the more you see exactly the same object. Moreover, fractals naturally have a dimension that is not an integer, not 1, not 2, but often somewhere in between, such as dimension 1.4176, whatever that means! We will discuss these concepts in more detail in Chapter 2.

Here are some examples of the types of images that we will study. In Plate 1 we show the Julia set of the simple mathematical expression $z^2 + c$, where both z and c are complex numbers. In this particular case, $c = -.122 + .745i$.

This image is called *Douady's rabbit*, after the French mathematician Adrien Douady whose work we will discuss in Chapter 2. The black region in this image resembles a "fractal rabbit." Everywhere you look, you see a pair of ears. In the accompanying figures, we have magnified portions of the rabbit, revealing more and more pairs of ears.

As we will describe later, the black points you see in these pictures are the "non-chaotic" points. They are points representing values of z that, under iteration of this quadratic function, eventually tend to cycle between three different points in the plane. As a consequence, the dynamical behavior in the black regions is quite predictable. All of this is by no means apparent right now, but by the time you have read Chapter 2, you will consider this example a good friend. Points that are colored in this picture also behave predictably: they are points that "escape"; that is, they tend to infinity under iteration. The colors here simply tell us how quickly a point escapes, i.e., how many iterations it takes to go beyond a pre-determined bound. Red points escape fastest, followed in order by orange, yellow, green, blue, and violet points. The boundary between these two types of behavior—the interface between the escaping and the cycling points—is the Julia set. This is where we will encounter all of the chaotic behavior for this dynamical system.

In Plates 2–5, we have displayed Julia sets for other quadratic functions of the form $z^2 + c$. Each picture corresponds to a different value of c. For example, Plate 5a is a picture of the Julia set for $z^2 + i$. As we see, these Julia sets may assume a remarkable variety of shapes. Sometimes the images contain large black regions as in the case of Douady's rabbit. Other times the Julia set looks like an isolated scatter of points, as in Plate 5b. Many of these Julia sets are *Cantor sets*. These are very complicated sets that arise often in the study of dynamics. We will begin our study of Cantor sets in Section 5 when we introduce the most basic fractal of all, the *Cantor middle-thirds set*.

All of the Julia sets in Plates 1–5 correspond to mathematical expressions that are of the form $z^2 + c$. As we see, when c varies, these Julia sets change considerably in shape. How do we understand the totality of all of these shapes, the collection of all possible Julia sets for quadratic functions? The answer is called the *Mandelbrot set*. The Mandelbrot set, as we will see in Section 24, is a dictionary, or picture book, of all possible quadratic Julia sets. It is a picture in the c-plane that provides us with a road map of the quadratic Julia sets. This image, first viewed in 1980 by Benoit Mandelbrot, is quite important in dynamics. It completely characterizes the Julia sets of quadratic functions. It has been called one of the most intricate and beautiful objects in all of mathematics. Amazingly, we still do not completely understand the structure of this set. That is, we still do not fully understand the dynamics of the simple quadratic function $z^2 + c$!

Plate 6 shows the full Mandelbrot set. Note that it consists of a basic central cardioid shape, with smaller bulbs or decorations attached. Plates 7–11 are magnifications of some of these decorations. Note how each decoration differs from the others. Buried deep in various regions of the Mandelbrot set, we also see small black regions which are actually small copies of the entire set. Look at the "tail" of the Mandelbrot set in Plate 7. The Mandelbrot set possesses an amazing amount of complexity, as illustrated in Plate 11 and its magnifications. Nonetheless, each of these small regions has a distinct dynamical meaning, as we will discuss in Section 24.

In this book we will also investigate the chaotic behavior of many other functions. For example, in Plates 12 and 13 we have displayed the Julia set for several functions of the form $c\sin(z)$ and $c\cos(z)$. Plate 15 displays the Julia sets for certain rational functions of the form $z^n + c/z^n$. These sets are what are known as "Sierpinski curves," probably the most interesting planar fractals, as we shall discuss in Section 25. If we investigate exponential functions of the form $c\exp(z)$, we find Julia sets that look quite different, for example, in Plate 16. And we can also look at the parameter planes (the c-planes) for these maps, a portion of which is shown in Plate 17.

In addition, we have referenced a number of online videos that are posted on my website that will allow you to see the dramatic changes these systems undergo as parameters vary. All of these videos are available at `math.bu.edu/DYSYS/animations.html`.

The images in this mathematical tour show quite clearly the great beauty of mathematical dynamical systems theory. But what do these pictures mean and why are they important? These are questions that we will address in the remainder of this book.

1.2 A Brief History of Dynamics

Dynamical systems has a long and distinguished history as a branch of mathematics. Beginning with the fundamental work of Isaac Newton, differential equations became the principal mathematical technique for describing processes that evolve continuously in time. In the eighteenth and nineteenth centuries, mathematicians devised numerous techniques for solving differential equations explicitly. These methods included Laplace transforms, power series solutions, variation of parameters, linear algebraic methods, and many other techniques familiar from the basic undergraduate course in ordinary differential equations.

There was one major flaw in this development. Virtually all of the analytic techniques for solving differential equations worked mainly for linear differential equations. Nonlinear differential equations proved much more difficult to solve. Unfortunately, almost all of the most important processes in nature are inherently nonlinear.

An example of this is provided by Newton's original motivation for developing calculus and differential equations. Newton's laws enable us to write down the equations that describe the motion of the planets in the solar system, among many other important physical phenomena. Known as the n-body problem, these laws give us a system of differential equations whose solutions describe the motion of n "point masses" moving in space subject only to their own mutual gravitational attraction. If we know the initial positions and velocities of these masses, then all we have to do is solve Newton's differential equation to be able to predict where and how these masses will move in the future.

This turns out to be a formidable task. If there are only one or two point masses, then these equations may be solved explicitly, as is often done in a freshman or sophomore calculus or physics class. For three or more masses, the problem today remains completely unsolved, despite the efforts of countless mathematicians over the past three centuries. It is true that numerical solutions of differential equations by computers have allowed us to approximate the behavior of the actual solutions in many cases, but there are still regimes in the n-body problem where the solutions are so complicated or chaotic that they defy even numerical computation.

Although the explicit solution of nonlinear ordinary differential equations has proved elusive, there have been four landmark events over the past 130 years that have revolutionized the way we study dynamical systems. Perhaps the most important event occurred in 1890. King Oscar II of Sweden announced a prize for the first scientist who could solve the n-body problem and thereby prove the stability of the solar system. Needless to say, nobody solved the original problem, but the great French mathematician Henri Poincaré came closest. In a beautiful and far-reaching paper, Poincaré totally revamped the way we tackle nonlinear ordinary differential equations. Instead of searching for explicit solutions to these equations, Poincaré advocated working qualitatively, using topological and geometric techniques, to uncover the global structure of all solutions. To him, a knowledge of all possible behaviors of the system under investigation was much more important than the rather specialized study of individual solutions.

Poincaré's prize-winning paper contained a major new insight into the behavior of solutions of differential equations. In describing these solutions, mathematicians had previously made the tacit assumption that what we now know as stable and unstable manifolds always match up. Poincaré questioned this assumption. He worked long and hard to show that this was always the case, but he could not produce a proof. He eventually concluded that the stable and unstable manifolds might not match up and could actually cross at an angle. When he finally admitted this possibility, Poincaré saw that this would cause solutions to behave in a much more complicated fashion than anyone had previously imagined. Poincaré had discovered what we now call *chaos*. Years later, after many attempts to understand the chaotic behavior of differential equations, he threw up his hands in defeat and wondered if anyone would ever understand the complexity he was finding. Thus, "chaos theory," as it is now called, really dates back over 130 years to the work of Henri Poincaré.

Poincaré's achievements in mathematics went well beyond the field of dynamical systems. His advocacy of topological and geometric techniques opened up whole new subjects in mathematics. In fact, building on his ideas, mathematicians turned their attention away from dynamical systems and toward these related fields in the ensuing decades. Areas of mathematics such as algebraic and differential topology were born and flourished in the twentieth century. But nobody could handle the chaotic behavior that Poincaré had observed, so the study of dynamics languished.

There were two notable exceptions to this. One was the work of the French mathematicians Pierre Fatou and Gaston Julia in the 1920s on the dynamics of complex analytic maps. They too saw chaotic behavior, this time on what we now call the *Julia set*. Indeed, they realized how tremendously intricate these Julia sets could be, but they had no computer graphics available to see these sets, and as a consequence, this work also stopped in the 1930s.

A little later, the American mathematician G. D. Birkhoff adopted Poincaré's qualitative point of view on dynamics. He advocated the study of iterative processes as a simpler way of understanding the dynamical behavior of differential equations, a viewpoint that we will adopt in this book.

The second major development in dynamical systems occurred in the 1960s. The American mathematician Stephen Smale reconsidered Poincaré's crossing stable and unstable manifolds from the point of view of iteration and showed by an example now called the "Smale horseshoe" (which we will cover in detail in Section 28) that the chaotic behavior that baffled his predecessors could indeed be understood and analyzed completely. The technique he used to analyze this is called *symbolic dynamics* and will be a major tool for us in this book. At the same time, the American meteorologist E. N. Lorenz, using a very crude computer, discovered that very simple differential equations could exhibit the type of chaos that Poincaré observed. Lorenz, who actually had been a Ph.D. student of Birkhoff, went on to observe that his simple meteorological model (now called the Lorenz system) exhibited what is called *sensitive dependence on initial conditions.* Roughly speaking, this means that a very small change in initial conditions in the system could lead to vastly different outcomes. Lorenz characterized this by saying that the flap of a butterfly's wings in Brazil could possibly set off a tornado in Texas. For him, sensitive dependence meant that long-range weather forecasting was all but impossible and showed that the mathematical topic of chaos was important in all other areas of science.

Then the third major development occurred in 1975 when T.Y. Li and James Yorke published an amazing paper called *Period Three Implies Chaos* [35]. In this paper, they showed that, if a simple continuous function on the real line has a point which cycles with period three under iteration, then this function must also have cycles of all other periods. Moreover, they also showed that this function must behave chaotically on some subset of the line. Perhaps most importantly, this paper was essentially the first time the word "chaos" was used in the scientific literature, and this motivated a huge number of mathematicians, scientists, and engineers to begin investigating this phenomenon.

Curiously, the Li-Yorke result was preceded by a research paper that had much more substantial results. In a 1964 paper [47], Oleksandr Sharkovsky determined that, if such a map on the real line had a cycle of period n, then he could list exactly all of the other periods that such a map must have. Unfortunately, this paper was published in Ukranian and hence was not known at all in the west. However, after the Li-Yorke paper appeared, this result became one of the most interesting results in modern dynamical systems theory. All of these results are described in Section 10.

These advances led to a tremendous flurry of activity in nonlinear dynamics in all areas of science and engineering in the ensuing decade. For example, the ecologist Robert May found that very simple iterative processes that arise in mathematical biology could produce incredibly complex and chaotic behavior. The physicist Mitchell Feigenbaum, building on Smale's earlier work, noticed that, despite the complexity of chaotic behavior, there was some semblance of order in the way systems became chaotic. Physicists Harry Swinney and Jerry Gollub showed that these mathematical developments could actually be observed in physical applications, notably in turbulent fluid flow. Later, other systems, such as the motion of the former planet Pluto or the beat of the human heart, have been shown to exhibit similar chaotic patterns. In mathematics, meanwhile, new techniques were developed to help understand chaos. John Guckenheimer and Robert F. Williams employed the theory of strange attractors to explain the phenomenon that Lorenz had observed a decade earlier. And tools such as the Schwarzian derivative, symbolic dynamics, and bifurcation theory—all topics we will discuss in this book—were shown to play an important role in understanding the behavior of dynamical systems.

The fourth and most recent major development in dynamical systems was the availability of high-speed computing and, in particular, computer graphics. Foremost among the computer-generated results was Mandelbrot's discovery in 1980 of what is now called the *Mandelbrot set*. This beautiful image immediately reawakened interest in the old work of Julia and Fatou. Using the computer images as a guide, mathematicians such as Bodil Branner, Adrien Douady, John Hubbard, Dennis Sullivan as well as Fields Medalists John Milnor, Bill Thurston, Curt McMullen, Artur Avila, and Jean-Christophe Yoccoz jumped into the field and greatly advanced the classical theory. Other computer graphics images such as the orbit diagram and the Lorenz attractor also generated considerable interest among mathematicians and led to further advances.

One of the most interesting side effects of the availability of high-speed computing and computer graphics has been the development of an experimental component in the study of dynamical systems. Whereas the old masters in the field had to rely solely on their imagination and their intellect, now mathematicians have an invaluable additional resource to investigate dynamics: the computer. This tool has opened up whole new vistas for dynamicists, some of which we will sample in this book.

Finally, as mentioned earlier, even the iteration of simple functions on the real line can behave quite chaotically. Indeed, for years the behavior of simple quadratic functions such as $x^2 + c$ or $kx(1 - x)$ when iterated was not completely understood. It was only with the appearance of the Mandelbrot set as well as with techniques from complex analysis that we finally understood this behavior in the 1990s. Think about it: We finally understood the iteration of quadratic functions on the real line in the 1990s! That being said, the behavior of cubic functions (and higher degree polynomials) is still not understood! Clearly, there are lots of open problems in the area of dynamical systems, many of which you will begin to see in this book. Please do solve several of these major open problems, and enjoy your Fields Medal!

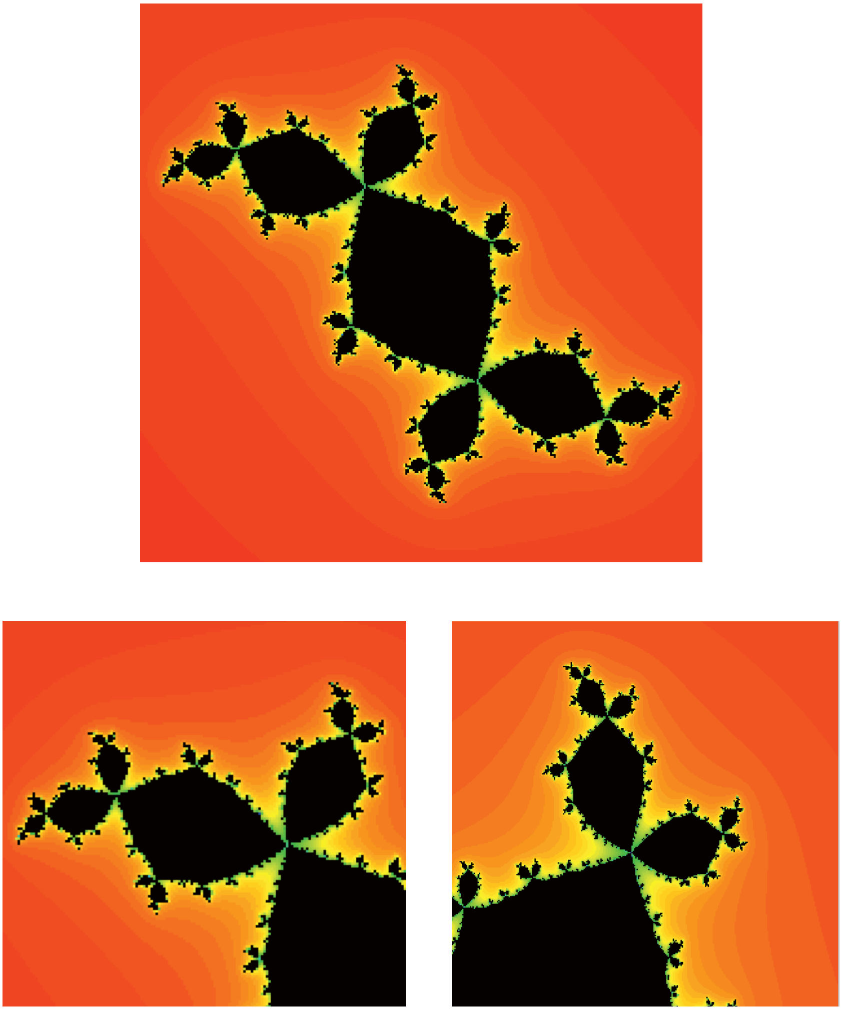


Plate 1: Douady's rabbit Julia set and several magnifications.

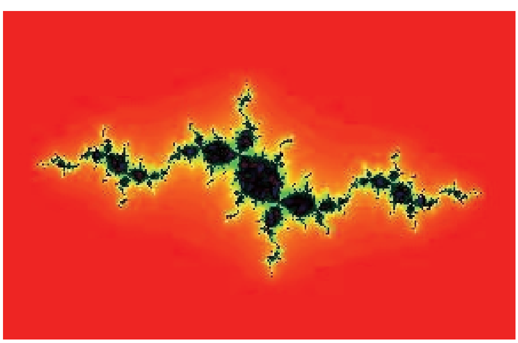


Plate 2: Dancing rabbits Julia set.

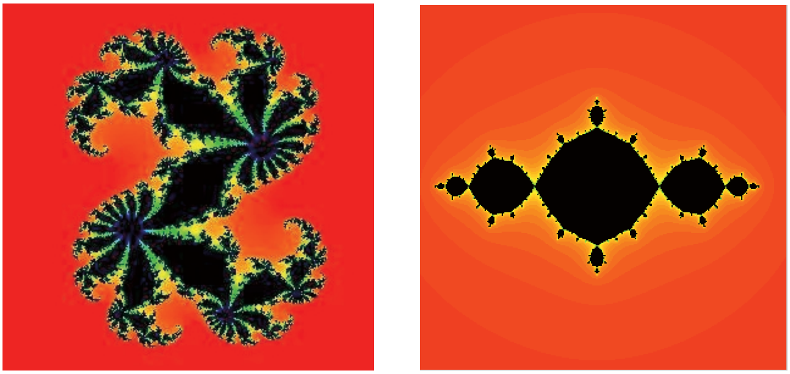


Plate 3: A dragon and the basilica Julia set.

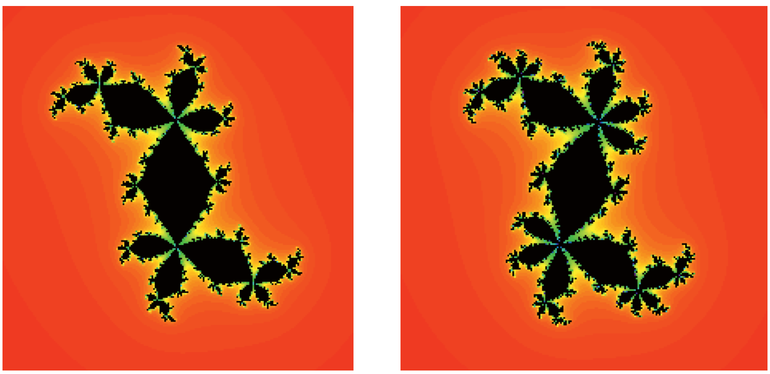
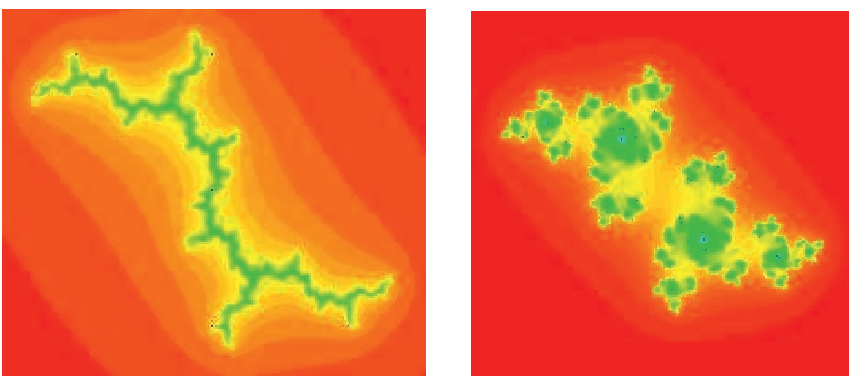


Plate 4: Three and four-eared rabbit Julia sets.



(a) (b)

Plate 5: A dendrite and a Cantor set Julia set.

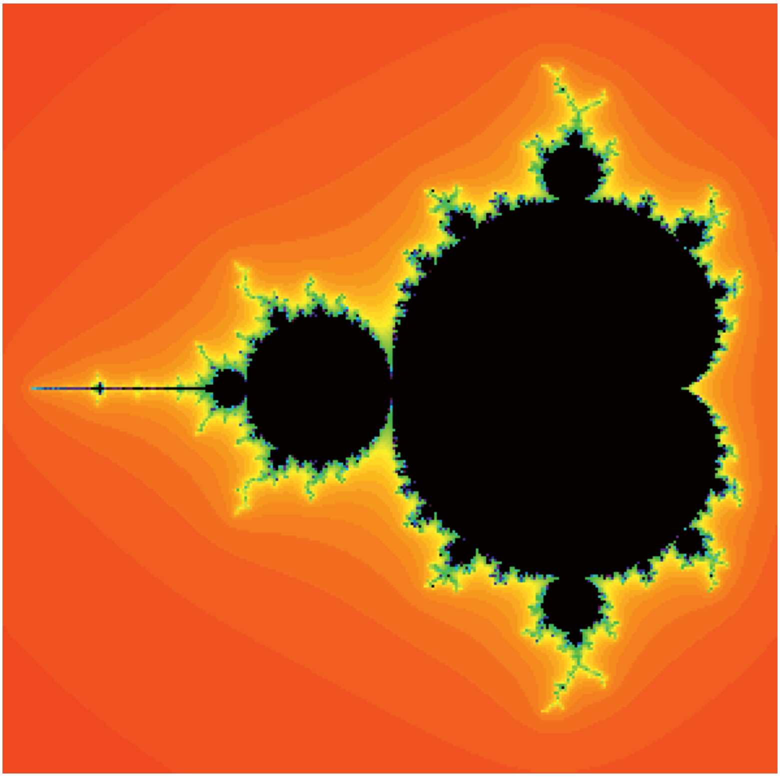


Plate 6: The Mandelbrot set.

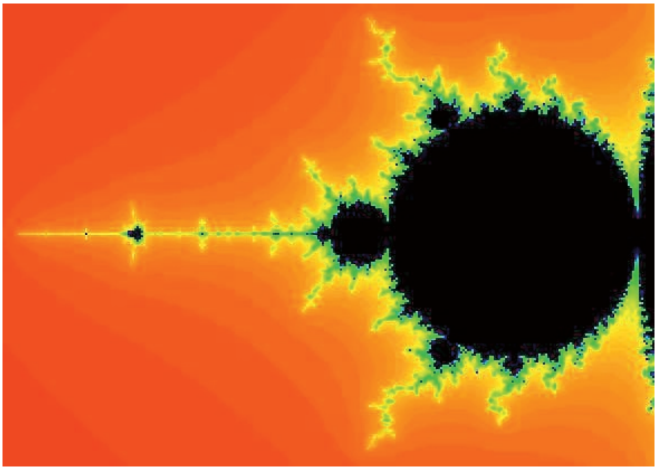


Plate 7: The tail of the Mandelbrot set.

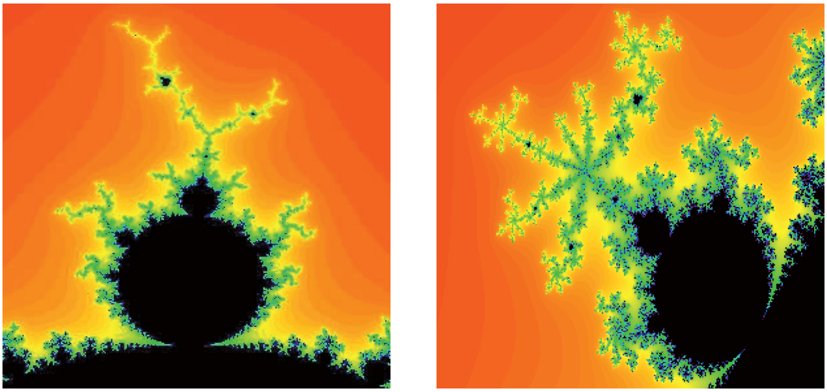


Plate 8: A period 3 and 7 bulb in the Mandelbrot set.

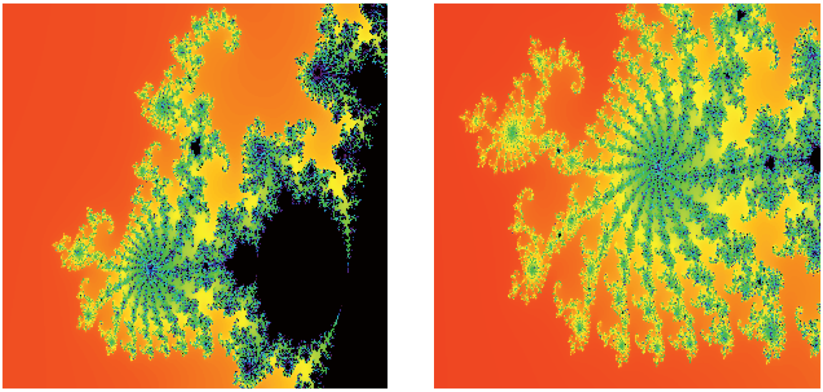


Plate 9: A period 17 bulb and a magnification.

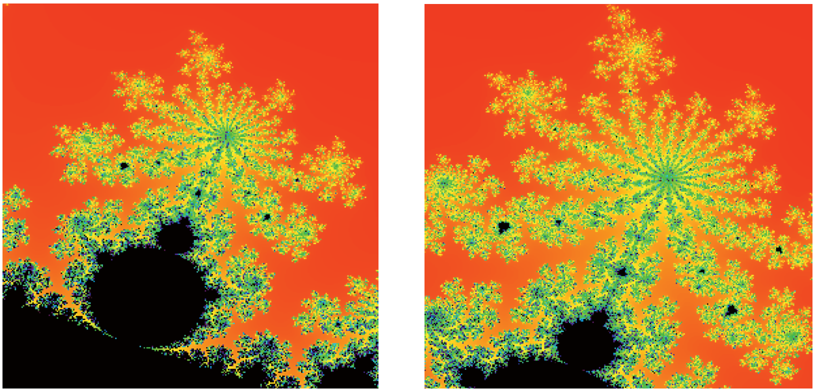


Plate 10: Another period 17 bulb and a magnification.

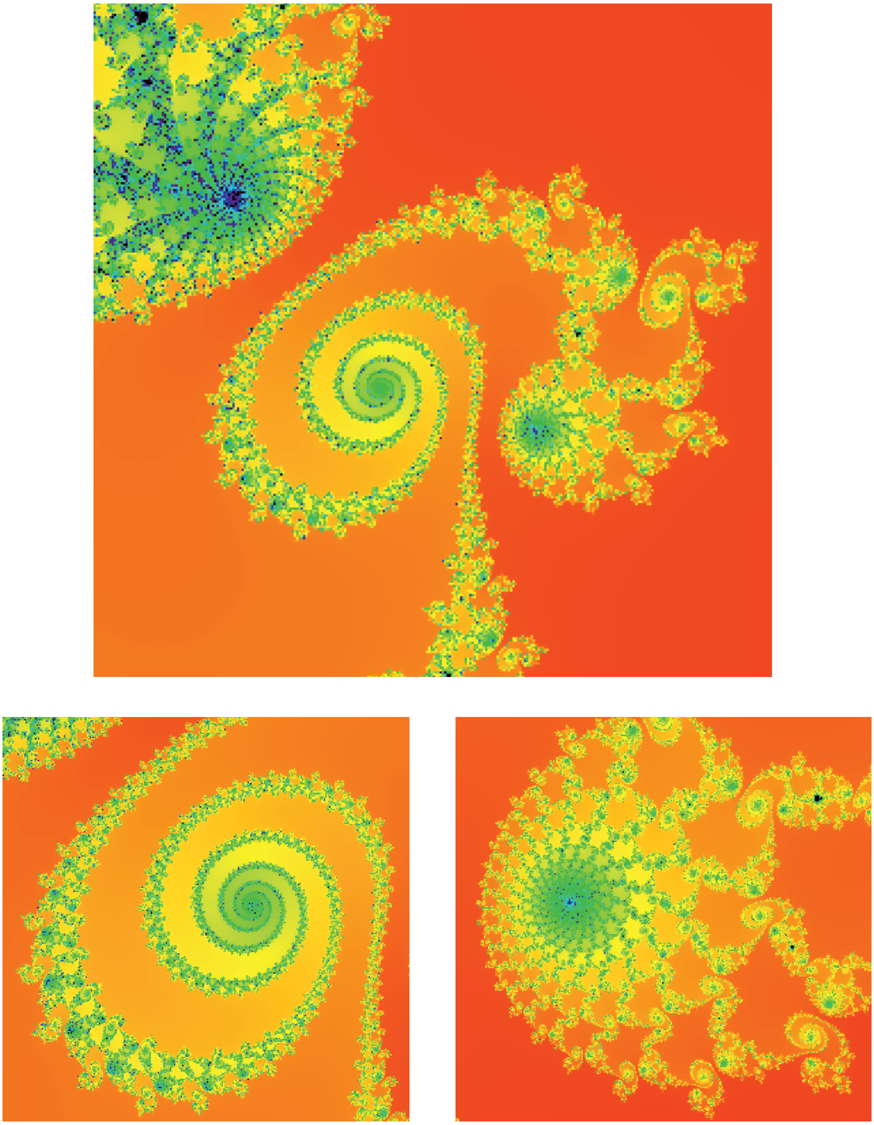
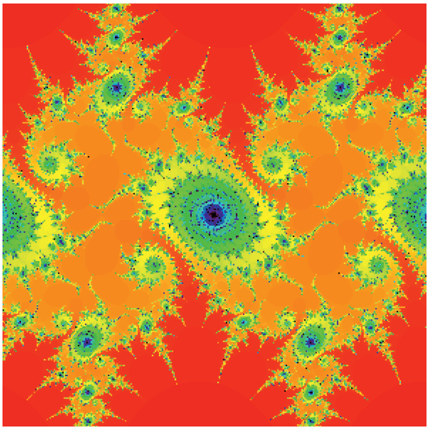


Plate 11: Another region in the Mandelbrot set.



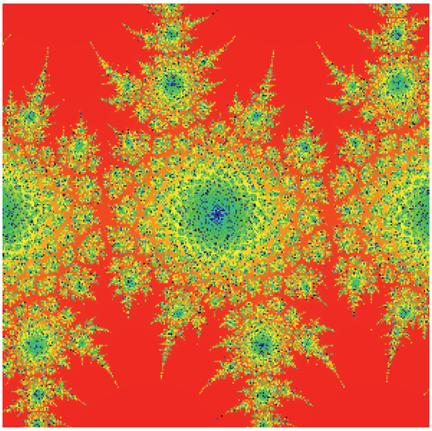


Plate 12: Julia sets of $(1+0.2i)\sin(z)$ and $(.61+.81i)\sin(z)$.

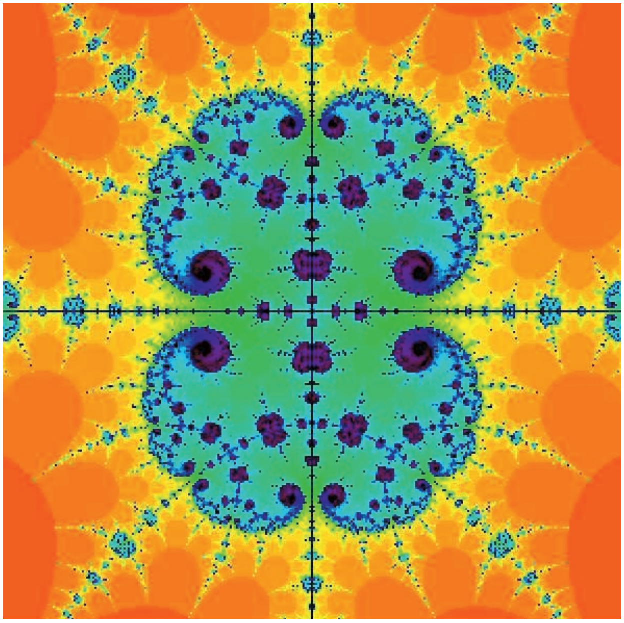


Plate 13: The Julia set of $2.95\cos(z)$.

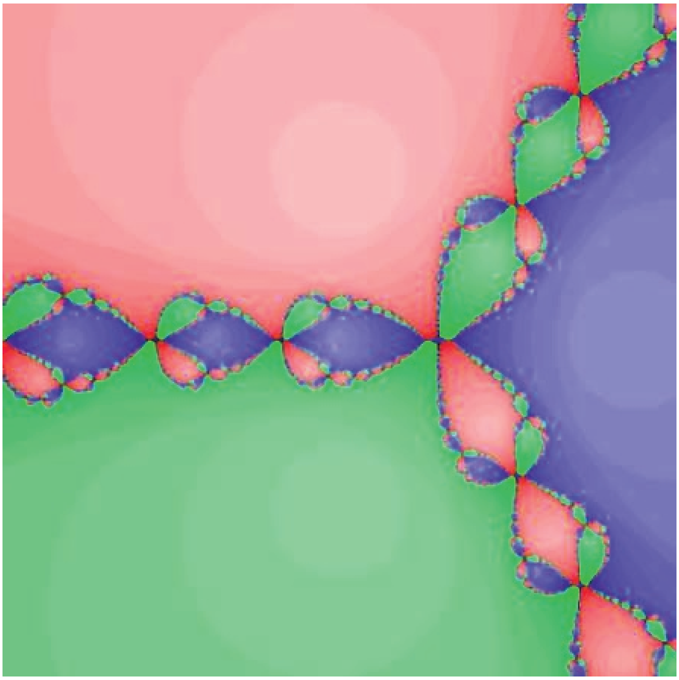


Plate 14: Complex Newton's Method for $z^3 - 1$.

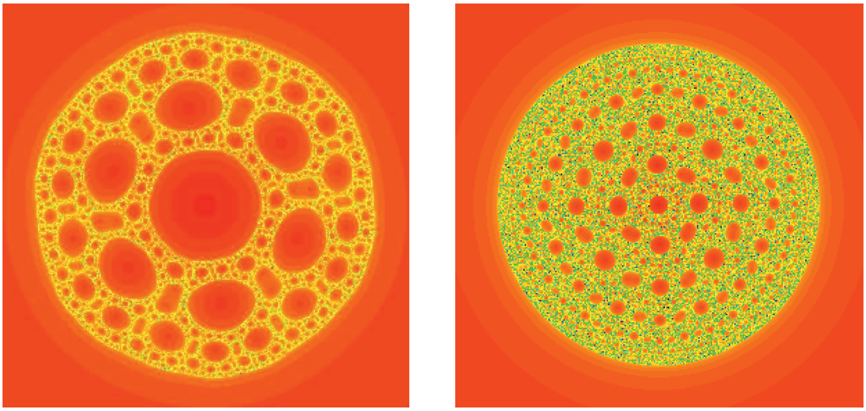


Plate 15: Sierpinski curve Julia sets for $z^3 + (-0.25 + .03i)/z^3$ and $z^2 - .004/z^2$.

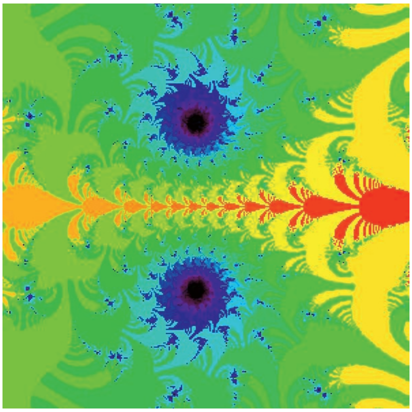



Plate 16: The Julia sets of $0.36\exp(z)$ and $0.38\exp(z)$ exhibiting a major "bifurcation."

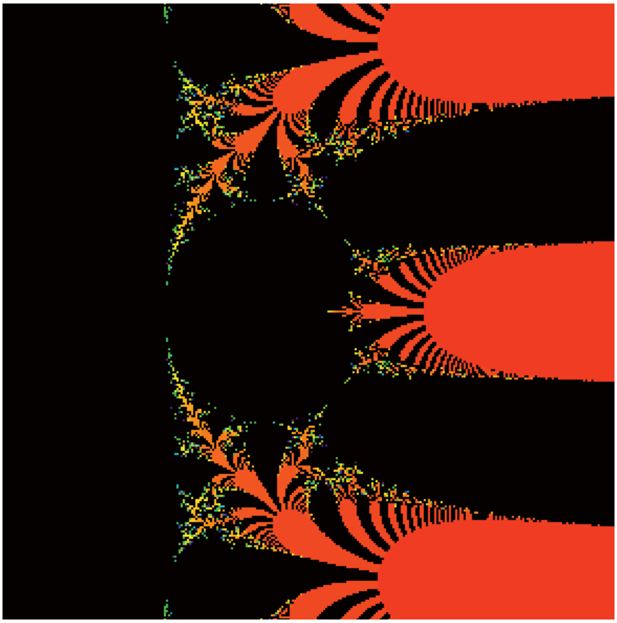


Plate 17: The parameter plane for $\lambda\exp(z)$.

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